Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences

Information Systems Department

Course Title:	ERP Systems
Course Code:	IS 451
Assessment	Project
Semester:	1st 2025-26
Marks:	20

Student Name (in English):	Student ID	Section No:
Lujain Alowais	443013844	371
Ghala Fahad Alsaleh	443011701	371
Reef Alotaibi	442014995	371
Layan Alhumaidi	443011837	371

Instructions:

1. Students should work in a group of **4-5 students** to solve this project.
2. Any delayed submission will not be accepted.
3. Any cheating work will be marked zero.

Official Use Only	
Student(s) Marks	Marks
	20

TABLE OF FIGURE

Figure 1 Inventory Management 5
Figure 2 Nahdi CRM 6
Figure 3 E-Commerce System 7
Figure 4 In-Pharmacy Purchase Process 8
Figure 5 Inventory Management(TO-BE) 9
Figure 6 Nahdi CRM (TO-BE) 10
Figure 7E-Commerce System (TO-BE) 11
Figure 8 In-Pharmacy Purchase Process(TO-BE) 12
Figure 9 Storage Locations Configuration 13
Figure 10 Putaway Rules Setup 13
Figure 11 Product Classification by Category 14
Figure 12 Goods Receipt Registration 14
Figure 13 Automatic Internal Transfer 15
Figure 14 Inventory Management dashboard 15
Figure 15Product Restocking Availability Display 16
Figure 16 E-Commerce Checkout Process 17
Figure 17 Complementary Product Display 18
Figure 18 Delivery Method Settings 18
Figure 19 Security Lead Time for Sales 19
Figure 20Purchase orders processed 19
Figure 21Scan the barcode while 20
Figure 22Payment and completion of the process. 21
Figure 23Invoice after payment. 21
Figure 24Creating branches 22
Figure 25Creating Storage Shelves 22
Figure 26Displaying the locations of the storage 23
Figure 27Distributing the Vitamin D product on shelves. 23
Figure 28Customer Complaint Submission Form 24
Figure 29Instant Submission Feedback 25
Figure 30Complaint Records & Status Overview 25
Figure 31Final Decision Notification to Customer 26

Contents

1. Introduction 4
2. As-Is Process Maps 4
3. To-Be Process Maps 8
4. Odoo Modules Implementation 12

1. Introduction

Nahdi Medical Company is one of the largest and most recognized pharmacy chains in Saudi Arabia, operating hundreds of branches across the kingdom and serving millions of customers every year. Its operations involve highly dynamic activities such as inventory management, procurement, warehousing, quality control, and continuous coordination between different departments. Because of the scale and complexity of these processes, Nahdi represents a realistic and meaningful case for studying enterprise-wide automation and digital transformation.

The decision to select Nahdi as the subject for this ERP project stems from the fact that it faces operational challenges common to large retail and healthcare organizations: high product turnover, sensitive storage requirements, strict quality standards, and the need for accurate and timely data across all branches. These challenges make Nahdi an ideal model for demonstrating how an ERP system can streamline workflows, reduce manual errors, and improve overall visibility across the supply chain.

By analyzing Nahdi's existing processes and redesigning them using an ERP solution such as Odoo, this project aims to highlight how technology can simplify operations, support data-driven decision-making, and enhance efficiency. The redesigned To-Be processes showcase how automation, system-driven validation, and intelligent forecasting can significantly improve inventory accuracy, reduce operational delays, and support proactive replenishment planning. As a result, Nahdi stands as a strong example of how ERP systems enable organizations to achieve higher levels of performance and operational excellence.

2. As-Is Process Maps

3.1 Inventory Management

The AS-IS process represents the current workflow used by Nahdi Pharmacy to manage inventory receiving, verification, recording, and reconciliation. This process relies heavily on manual activities performed by warehouse staff, inventory controllers, and the finance department.

The workflow begins when warehouse staff receive incoming goods and conduct a manual quality check. Based on this check, items are either approved or rejected. If approved, the staff manually record the receipt in the ERP system.

Once the items are entered in the system, the Inventory Controller updates stock records, assigns storage locations, and manually generates an inventory report. The report often requires validation, and if errors are found, the controller must revise the data before continuing. After validating the report, the controller carries out a physical stock count, compares results against the system, and determines whether the stock is placed in the correct location. If incorrect, a manual stock transfer is performed.

At the end of the cycle, the Finance Department receives reconciliation reports and compares them against system data before adjusting financial records. This version of the process is highly dependent on human effort, susceptible to delays or inconsistencies, and lacks automated decision-making, forecasting, or system-driven checks.

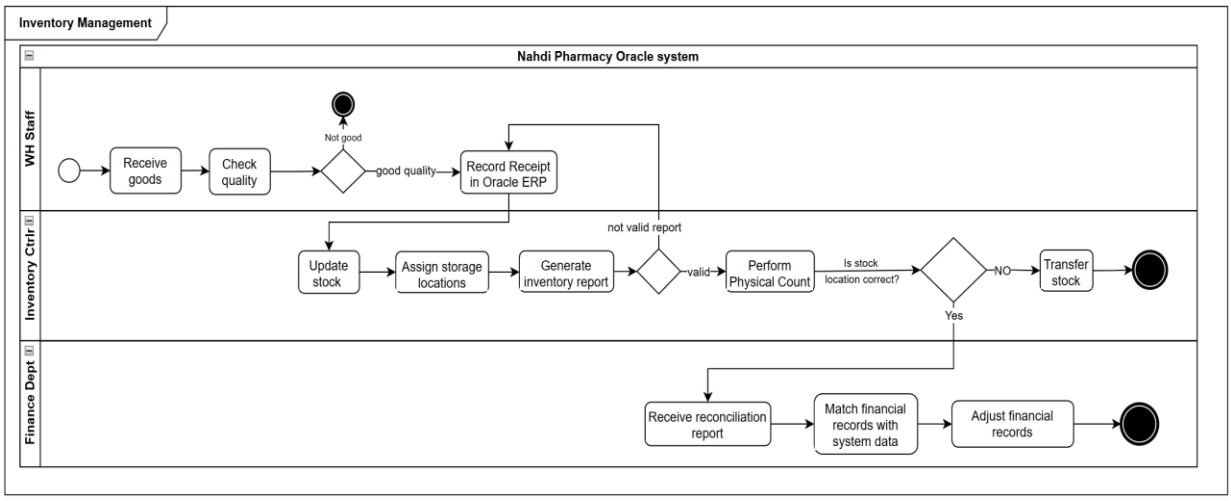


Figure 1 Inventory Management

3.2 Nahdi CRM Process:

Nahdi's existing practice for handling complaints is done manually and relies on various departments. When the order is first received by the customer and any issue is identified, they call the Customer Service Agent on phone. The agent checks customer authentication, opens a complaint ticket and requests information via email. After the customer manually provides the evidence, the Quality Team reviews the submitted

information and assesses the validity of the complaint. If the case is ok, the Quality Team raises a request for them to the Finance Team. Customer Service gets confirmation of completion and refund is processed upon verification of all details by Finance. The agent then notifies the customer and closes the ticket. If the complaint is not valid the Quality Team sends a rejection decision to the agent who will inform the customer and close the case. To sum up, AS-IS process communication happens manually, creates multiple handoffs and the resolution gets delayed as it involves Customer Service, Quality and Finance.

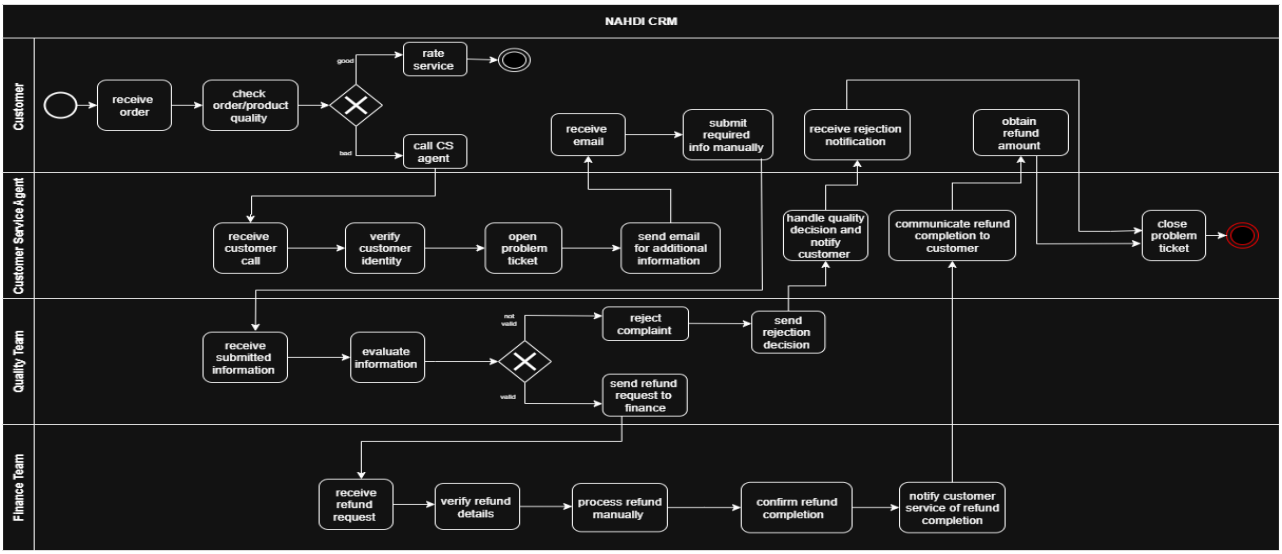


Figure 2 Nahdi CRM

3.3 E-Commerce System:

The AS-IS process represents the current workflow used to fulfill orders through the website, as this model is based on direct interaction from the customer with the system, with many steps requiring manual input or continuous confirmation from the user. The process begins when the customer browses the site and searches for the desired product. The system verifies the availability of the product, and if it is not available, the customer does not complete the purchase process. If the product is available, the customer can choose it and add it to the shopping cart. The customer then moves on to choose the shipping method, then logs in or creates a new account. When moving to the payment stage, the customer enters the required shipping information, while the system verifies the accuracy of the basket data and calculates the final cost, which includes VAT and

shipping fees. At this stage, the customer confirms the payment process, and the system verifies that the payment has been accepted from the service provider. In case of rejection, the customer needs to retry payment or use a different method. If payment is accepted, the system generates the order number and sends it to the warehouse for further processing. In the final part of the process, the shipping team receives the order, prepares the package, and delivers it to the customer who successfully receives their order. In general, the AS-IS process is characterized by its heavy reliance on manual input from the client, and its lack of smart or automated steps, which can lead to slow processing and a high probability of error.

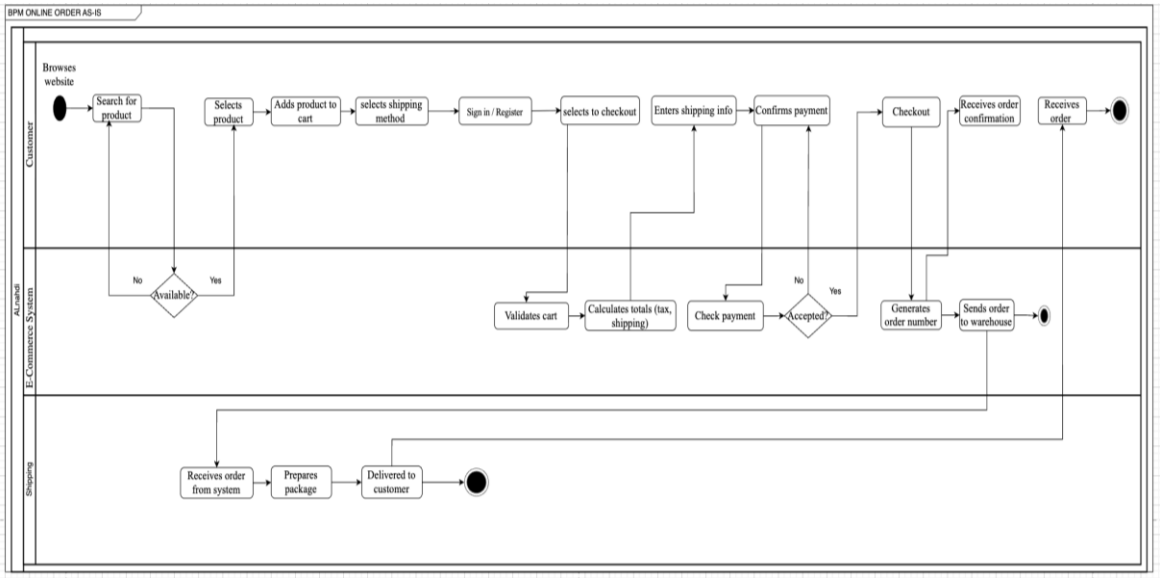


Figure 3 E-Commerce System

3.4 In-Pharmacy Purchase Process

In the AS-IS process in pharmacy purchasing process at Nahdi Pharmacy the customer begins by entering the pharmacy and waiting in line if they have a prescription or taking the product themselves if they don't. They then speak with the pharmacist who checks the order and partially searches the shelves for the medication which can take some time.

If the product is unavailable the customer is informed immediately but If it is available the pharmacist prepares the medication and prints the instructions manually.

The customer then proceeds to the cashier where the product is scanned manually the order details are displayed and payment is made and the order is handed to the customer. This process can be time-consuming for the customer.

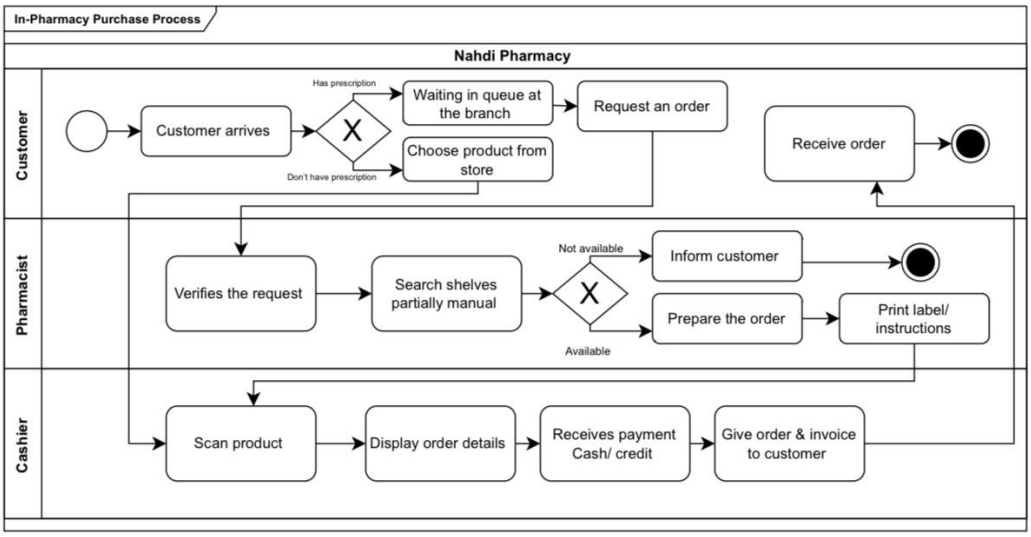


Figure 4 In-Pharmacy Purchase Process

3. To-Be Process Maps

3.1 Inventory Management

The TO-BE process illustrates the redesigned, optimized workflow enhanced by Odoo's automation capabilities, advanced inventory logic, and integrated analytics. This improved process reduces manual involvement, minimizes errors, and introduces smart decision-making features.

The process begins when warehouse staff scan incoming goods, triggering the system to perform an automated quality analysis. Based on smart rules, the system classifies items as approved or rejected. Approved items are auto-registered in Odoo's inventory without manual data entry.

Next, the Inventory Controller oversees an automated series of system-driven checks. Odoo evaluates the product's attributes (dimensions, category, turnover rate, FEFO/FIFO rules) and determines the optimal storage location. If a suitable location is found, the system auto-assigns it; otherwise, it sends an alert for manual assignment.

After the items are placed, the controller performs a smart physical count, and Odoo automatically validates the counted quantity, updates the final stock level, and adjusts

discrepancies. The system then analyzes historical consumption data and predicts the stock depletion date.

A decision gateway checks whether the stock level is expected to fall below the safety threshold. If so, the system auto-generates a replenishment recommendation and drafts a purchase order. If not, the system simply logs that the stock level is stable.

Finally, the Finance Department matches financial records with system data and updates accounting entries accordingly. The redesigned process is highly automated, significantly faster, and includes forecasting, auto-replenishment, and real-time visibility reflecting a modern ERP-driven inventory experience.

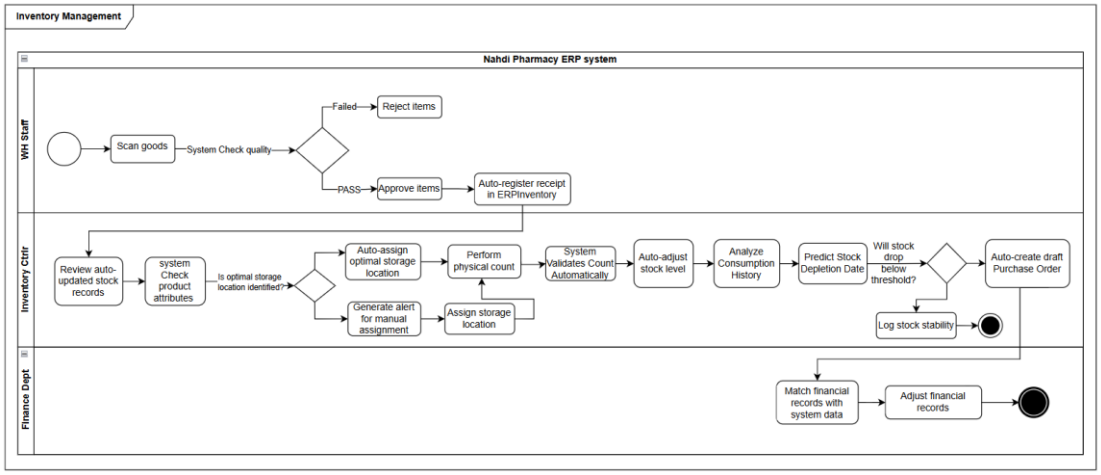


Figure 5 Inventory Management(TO-BE)

3.2 Nahdi CRM Process:

The TO-BE model projects a simplified complaint process that is completely digitized and automated through Nahdi CRM, unlike the current process which is manual and requires the involvement of multiple departments. Now, the customer need not call Customer Service but can submit a complaint directly through the application. They specify the order number, state their complaint and upload evidence. Once submission in CRM system, complaint is auto-received, validates data and classifies issue. If the complaint fits the automatic refund criteria, the refund amount gets calculated, the refund is processed instantaneously, the customer is notified and the case is closed without human intervention. If the complaint is not auto-resolvable, the CRM will route the case to the Quality Team. Reviewing submitted information quality assesses the accuracy of the complaint returning a decision respective of approval or rejection to the

system. The CRM executes the approved solution or sends a refusal notification to the customer. At last, case status is updated, and workflow is completed. The TO-BE process minimizes waiting time, avoids needless handoff, prevents involvement of Finance, lessen involvement of Customer Service faster complaint handling, accurate complaint handling & automated complaint handling.

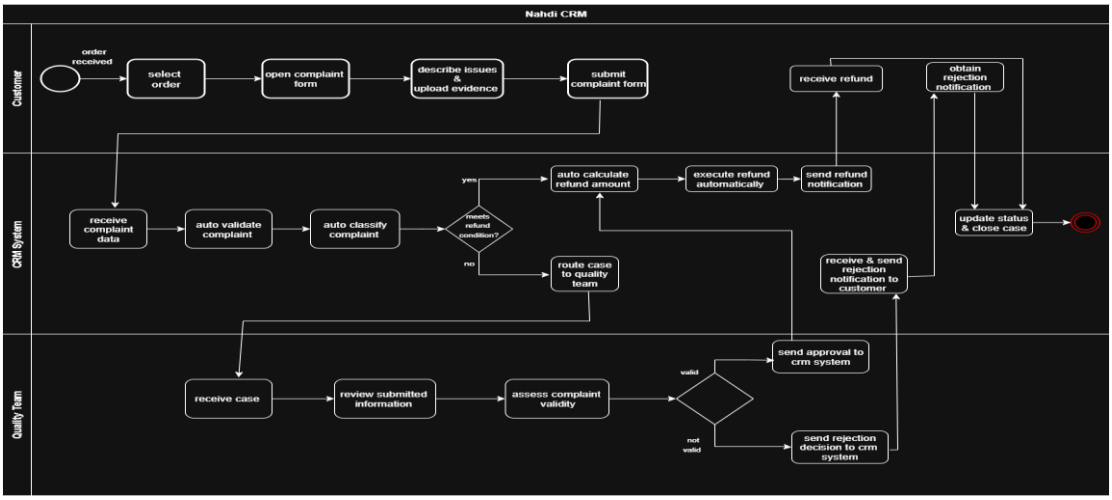


Figure 6 Nahdi CRM (TO-BE)

3.3 E-Commerce System:

The TO-BE model showcases an improved workflow based on greater automation and smarter steps to improve the customer experience and speed up the purchasing process. The model aims to reduce manual intervention, enhance data accuracy, and provide a seamless usage journey that is automatically driven by the system.

The process begins when the customer browses the site and selects the desired product. After adding the product to the cart, the system automatically displays detailed information including expected shipping dates. The system also introduces a new feature based on embedded intelligence, where it displays complementary or suggested products based on the nature of the chosen product for example:

When choosing a shampoo, the system suggests a suitable conditioner.

If the customer agrees to the proposed product, it is automatically added to the basket; if he does not agree, the process continues normally.

The customer then chooses the shipping method, logs in or creates a new account when needed. The customer then goes to the payment page and enters the shipping

information, while the system automatically verifies the authenticity of the basket and accurately calculates the final cost, including tax and shipping fees.

After completing the step, the customer confirms the payment process, and the system immediately verifies the payment process. If rejected, the system provides options to retry or use an alternative method. When payment is accepted, the system completes the process by generating the order number and sending it directly to the warehouse.

The shipping department receives the order immediately, begins preparing the package and setting a delivery date, while the customer receives instant notifications confirming the order, shipping status, and pickup time.

The new TO-BE model offers a more advanced experience based on automation, intelligent verification, and display of complementary products, which reduces errors and contributes to speeding up the process and providing a more effective and quality purchasing experience.

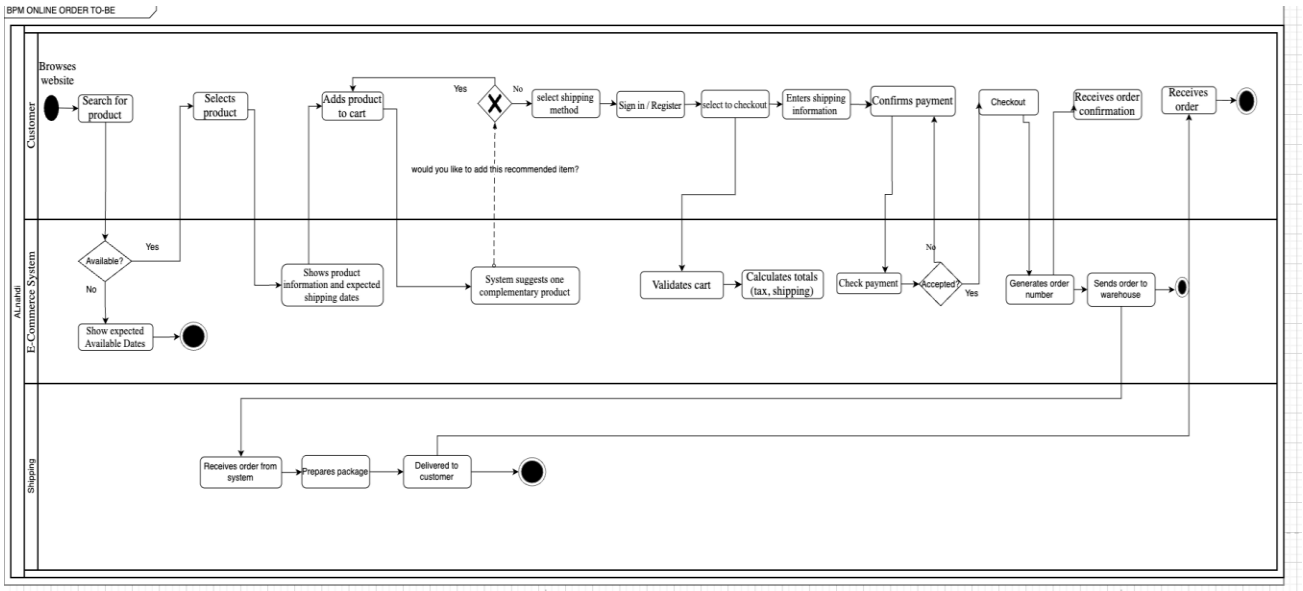


Figure 7E-Commerce System (TO-BE)

3.4 In-Pharmacy Purchase Process

In TO-BE the workflow will become faster and more accurate. Upon entering the customer selects the product or if they have a prescription and then click on the queueing machine to receive a number and then the customer will receive a number . The pharmacist then verifies the order and enters the data directly into the pharmacy

system which automatically checks product availability and displays its precise location on which shelf it will be found.

If the product is unavailable the system displays the nearest branch that carries it and notifies the pharmacist and then informs the customer.

If the product is available the system displays the location and automatically prints the medication label with the patient's name and dosage and expiration date without any manual intervention.

At the checkout the customer scans the barcode at the self-service cashier machine and the system displays the order details and records the payment and the customer can choose to receive the invoice by email or print it or both of this to choices and after the transaction is completed the customer receives their order.

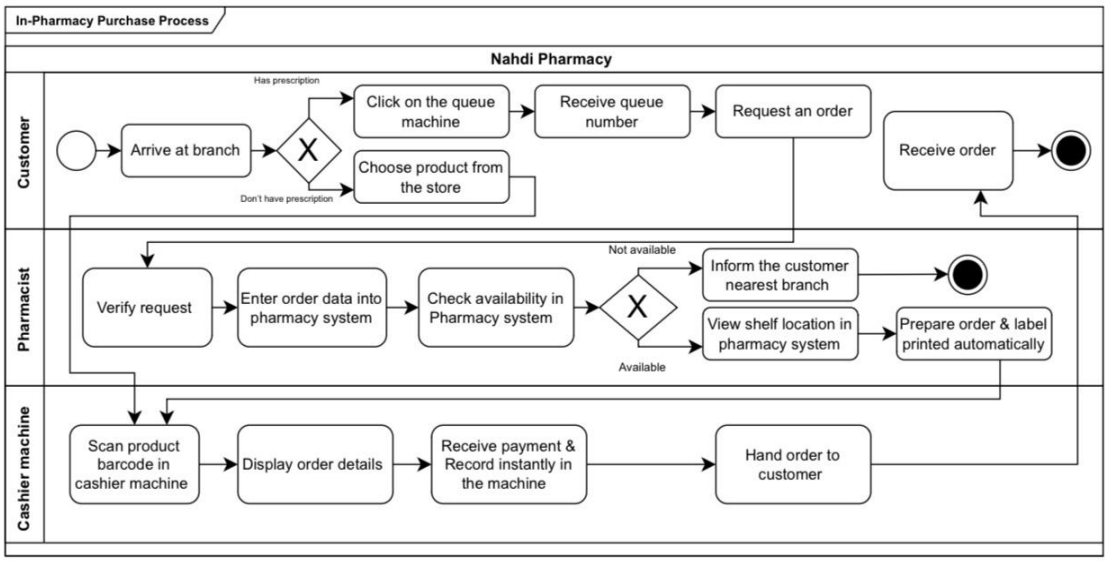


Figure 8 In-Pharmacy Purchase Process(TO-BE)

4. Odoo Modules Implementation

4.1 Inventory Management

4.1.1 Storage Locations Configuration

This figure shows the internal warehouse locations configured in Odoo, including Stock, Fast Moving, Refrigerated, and Expired areas, each assigned with an appropriate storage category to support automatic putaway and inventory control.

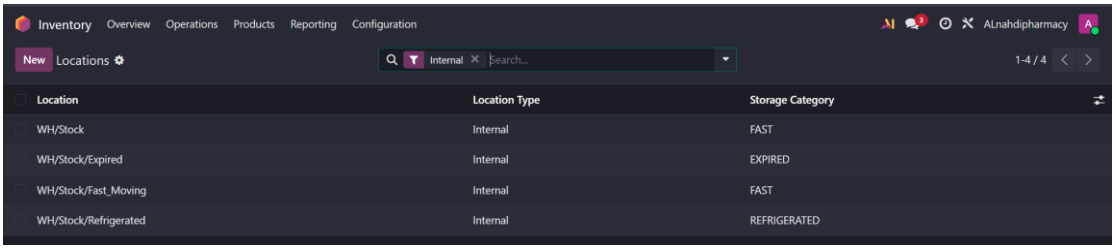


Figure 9 Storage Locations Configuration

4.1.2 Putaway Rules Setup

This figure shows the configured putaway rules that automatically route incoming products from the main stock location to their appropriate storage areas based on product category, including Fast Moving, Refrigerated, and Expired drugs.

The putaway rules were configured to automate the storage assignment process upon product receipt. When goods arrive at the main stock location, the system automatically redirects them to the correct sub-location based on their category, ensuring compliance with storage requirements and improving warehouse efficiency while reducing manual intervention and errors.

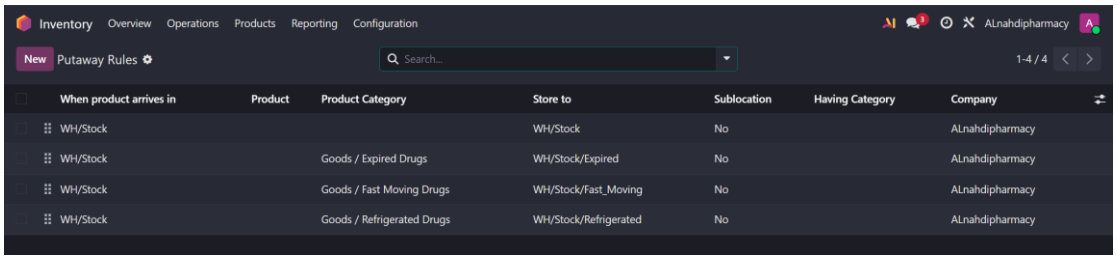


Figure 10 Putaway Rules Setup

4.1.3 Product Classification by Category

This figure illustrates the product classification mechanism in the Odoo Inventory system, where each product is assigned to a specific category. This classification enables automated storage routing, stock control, and reporting, and serves as the basis for applying putaway and replenishment rules.

This figure shows the classification of a pharmaceutical product (Panadol Extra Tablets) under the “Fast Moving Drugs” category in the Odoo Inventory system. This classification supports automated storage allocation, stock monitoring, and replenishment decision-making.

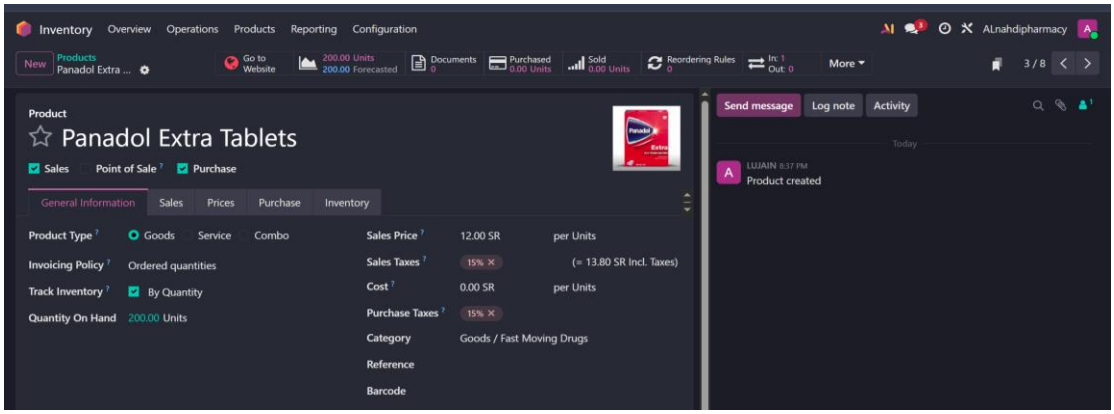


Figure 11 Product Classification by Category

4.1.4 Goods Receipt Registration

Once goods arrive from the supplier, the warehouse staff registers the receipt in the ERP system by creating a Goods Receipt document. The system records the supplier name, source purchase order, destination storage location, and received quantities. After verification, the receipt is validated and marked as "Done", which automatically updates the inventory stock levels in real time and assigns the goods to their designated storage locations.

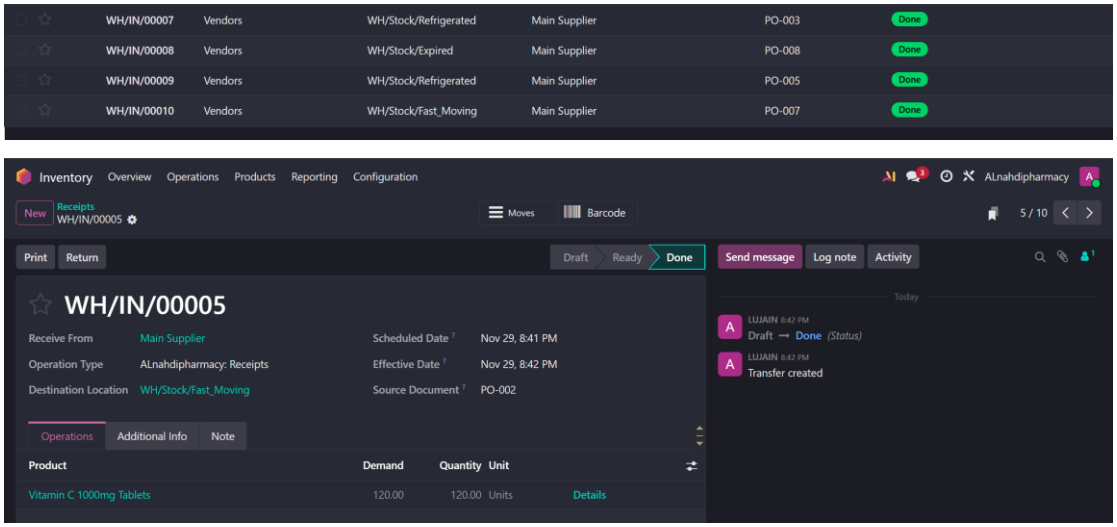


Figure 12Goods Receipt Registration

4.1.5 Automatic Internal Transfer

Automatic Internal Transfer is used to automatically move products between internal storage locations based on system rules and product status. In the implemented ERP system, when a product is identified as expired, the system automatically generates an internal transfer from the main warehouse location (WH/Stock) to the expired goods

location (WH/Stock/Expired) without manual intervention. This process ensures real-time inventory accuracy, proper segregation of expired items, and compliance with pharmaceutical storage regulations.

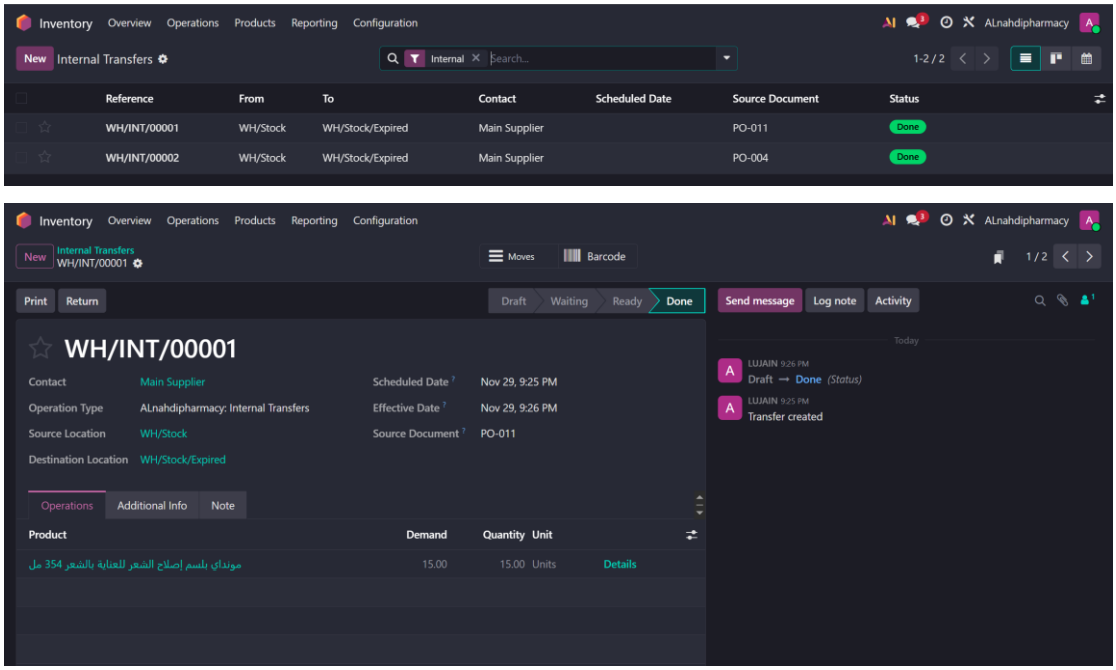


Figure 13 Automatic Internal Transfer

4.1.6 Inventory Management dashboard

This screen shows the **Inventory Management dashboard in the Odoo system** for the pharmacy. It displays a list of products with key stock information, including unit cost, total value, quantity on hand, available quantity, incoming and outgoing units. The interface also provides quick access to **product history, replenishment actions, and storage locations**, enabling efficient monitoring and control of inventory levels and supporting accurate stock planning and decision-making.

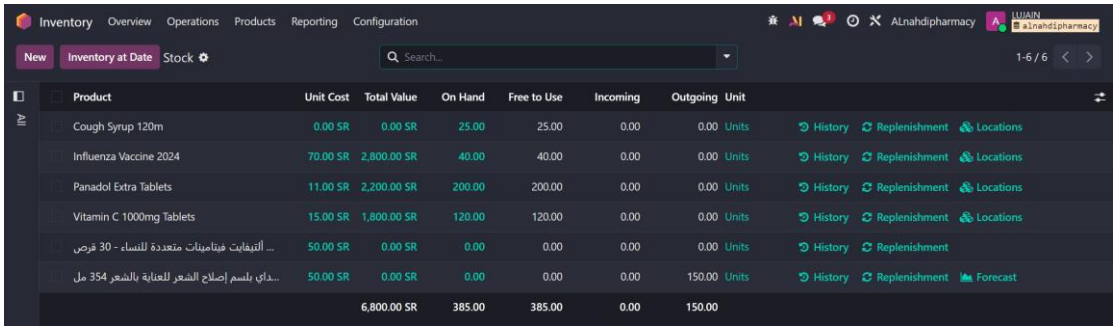


Figure 14 Inventory Management dashboard

4.2 E-Commerce System Order

4.2.1 Product Restocking Availability Display

This figure shows how Odoo displays the availability of a product when it is out of stock. The system clearly indicates that the item is currently unavailable and shows the expected restocking date. This helps users know when the product will be available again and makes it easier to plan and communicate with customers. The restocking date appears automatically based on the information updated in the system, which ensures that the date shown is accurate and up to date.

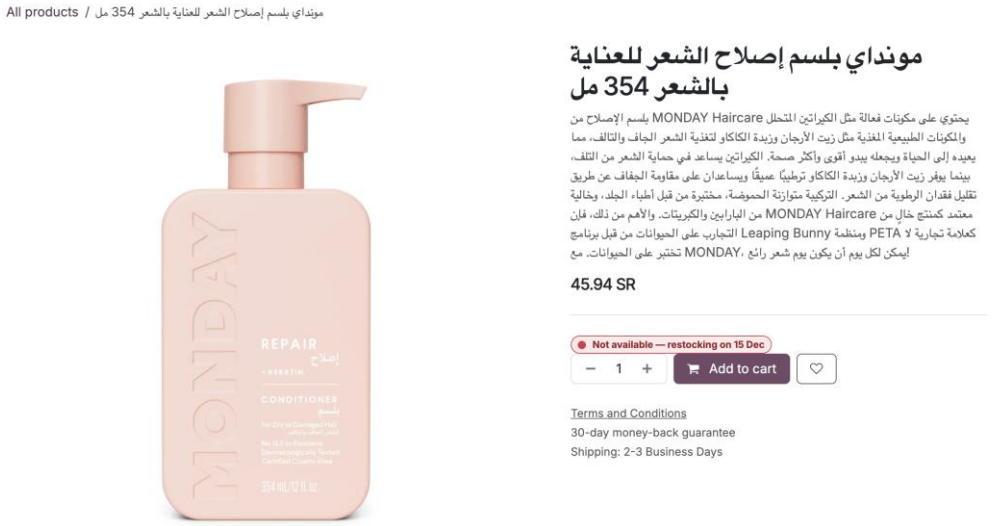


Figure 15Product Restocking Availability Display

4.2.1 E-Commerce Checkout Process

This figure shows part of the online store's payment interface, specifically the order summary that appears to the customer before completing the purchase. The system clearly displays all products added to the basket, in addition to the shipping fee recorded under Delivery Fee, which is 20 riyals. Total prices, taxes, and the final amount payable are also displayed.

An important point in this section is that a custom shipping service has been added, as the free version of Odoo does not directly support shipping. So, we created and added a shipping service manually in a way that ensures fees appear on the cart page and payment is organized and transparent to the customer.

This clear summary helps improve the user experience by giving them a complete picture of order costs before moving on to the checkout step, reducing confusion or inquiries regarding fees.

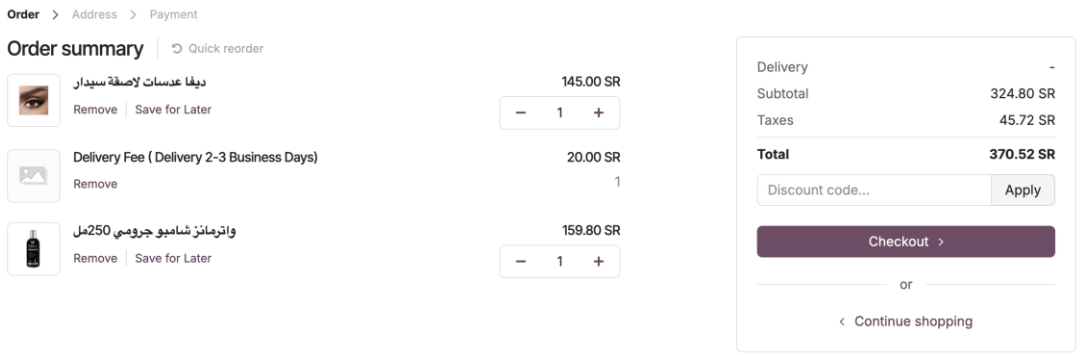


Figure 16 E-Commerce Checkout Process

4.2.3 Complementary Product Display

This figure shows how Odoo suggests complementary products when selecting an item. In this example, when the user adds contact lenses to cart, the system automatically displays an additional option for lens solution.

This feature helps the user see related items that they might need with the main product, which makes the shopping process easier and more convenient. It also improves the overall customer experience by grouping products that are commonly used together.

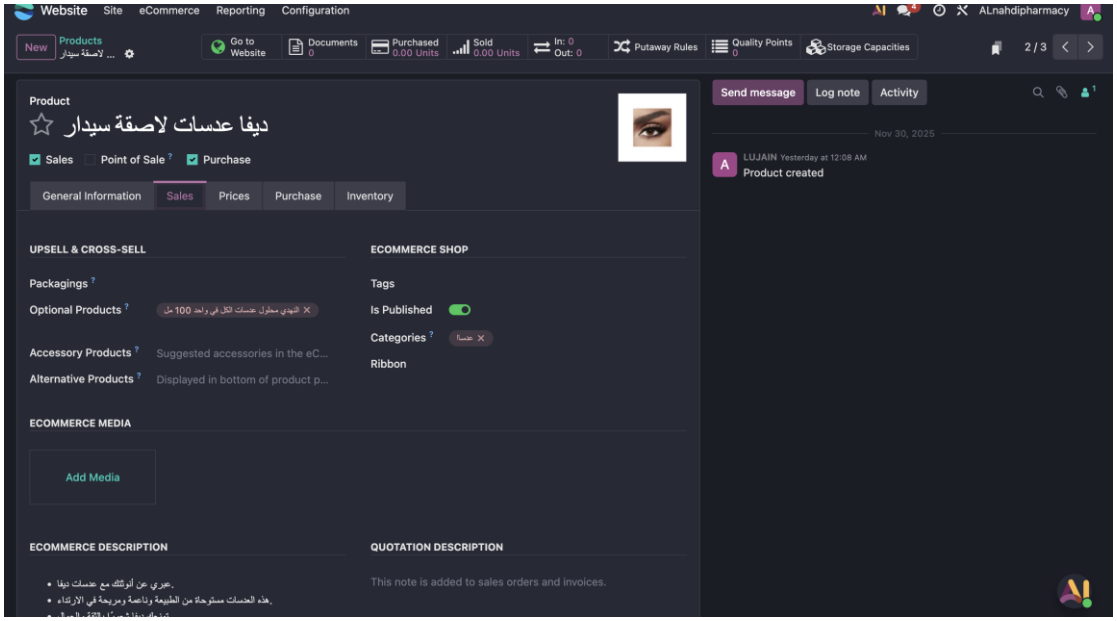


Figure 17 Complementary Product Display

4.2.4 Delivery Method Settings

This figure shows the Standard Delivery shipping method settings within the Odoo system, which specify the shipping details that later appear to the customer during the ordering process. In the settings shown, the service provider type is selected as Fixed Price with a fixed fee of SAR 20 applied, in addition to activating the free shipping option for orders exceeding SAR 250. The image also shows the product associated with the shipping process and the permitted country (Saudi Arabia), in addition to restrictions related to weight and size. These settings ensure that shipping costs are automatically and accurately calculated for each order and help adjust shipping policies to match store requirements.

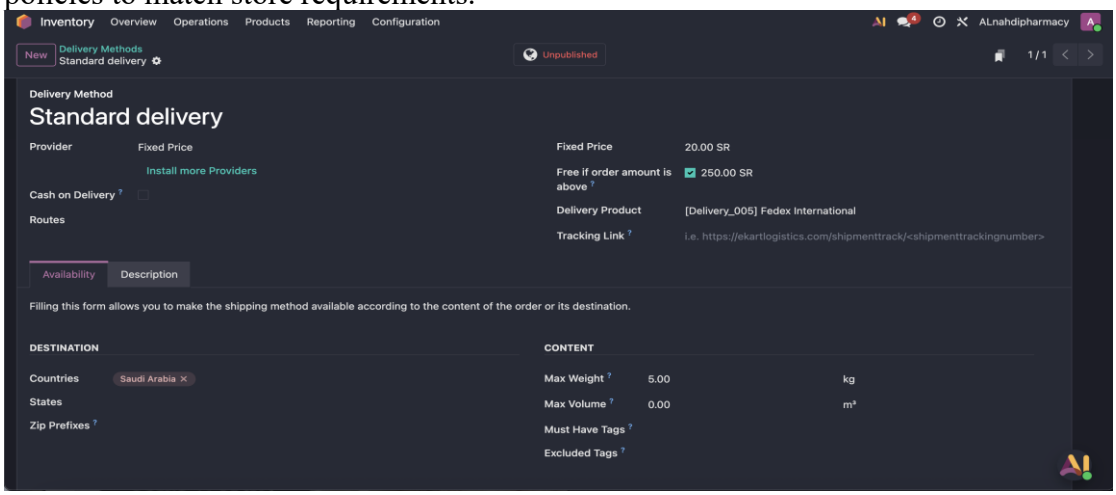


Figure 18 Delivery Method Settings

This figure shows the Security Lead Time for Sales setting within the system's repository settings. This setting is used to add a margin of safety to expected

delivery dates, as the system is set to advance the delivery date 10 days earlier than the actual date. The goal of this setup is to reduce the possibility of delays and provide a backup time for processing orders, especially in the event of any delays in the supply chain. This setup helps improve the accuracy of delivery times offered to customers and increase the reliability of the shipping service.

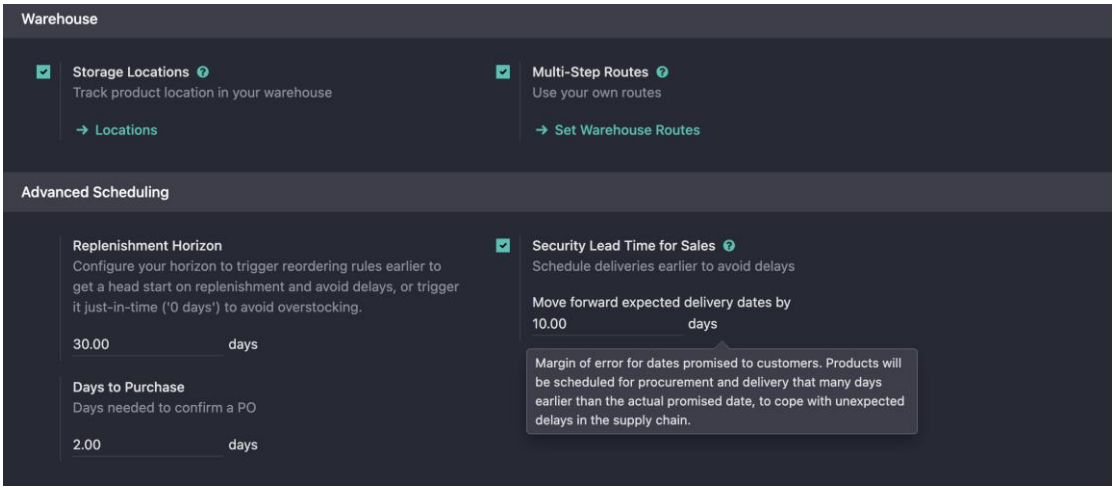


Figure 19 Security Lead Time for Sales

Purchase orders processed

This screen shows the Purchase Orders module in the Odoo system for Alnahdi Pharmacy. It displays a list of e-commerce purchase orders with key transactional details, including the reference number, confirmation date, vendor, buyer, total amount, billing status, and expected arrival time. The interface also provides quick access to order activities, source documents, and status updates, enabling efficient tracking and verification of all online orders. This view supports clear monitoring of order processing, ensures accurate billing, and helps maintain smooth and timely fulfillment across the pharmacy's e-commerce operations.

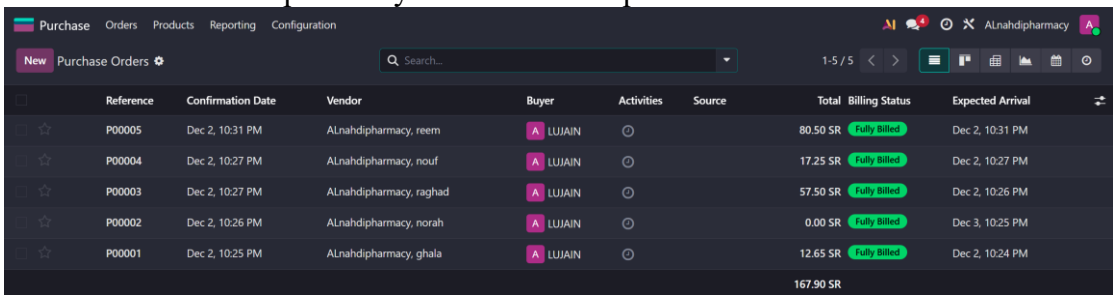


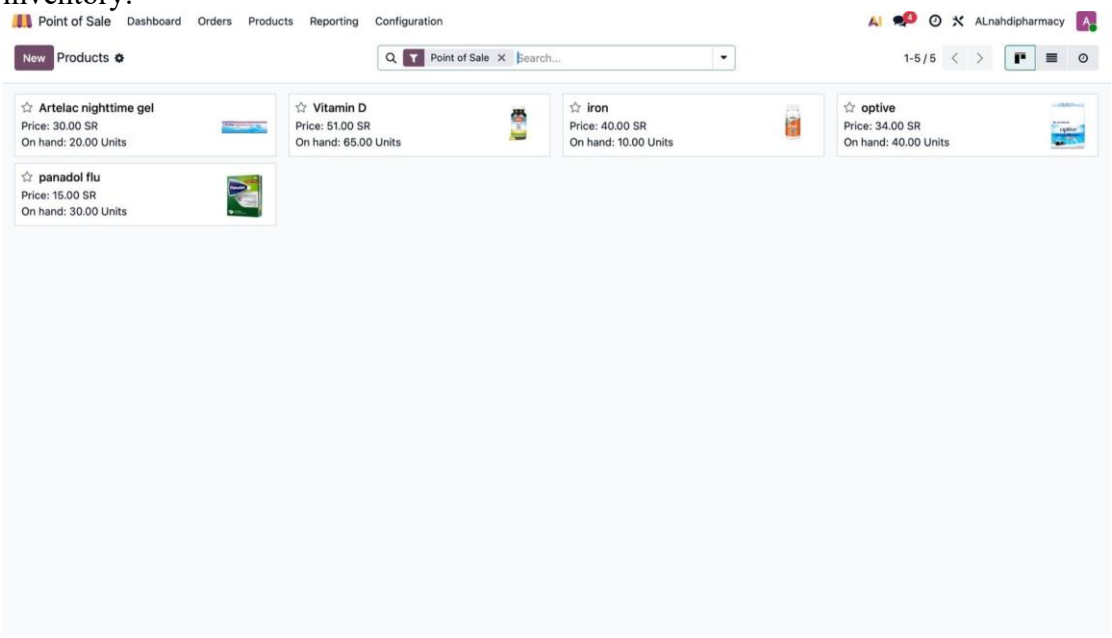
Figure 20Purchase orders processed

4.3 In-Pharmacy Purchase Process

4.3.1 Adding products to the ODOO system

After adding products a product list will appear in the Odoo POS system and showing the added product and its price and the quantity available at each branch this will

reflects the readiness for sale and ensures that each product is matched to the correct inventory.



4.3.2 Scan the barcode while purchasing using the self-service cashier machine.

The customer scans the barcode directly through the device's camera and the product automatically appears in the shopping cart within the POS system simulating a self-service checkout.

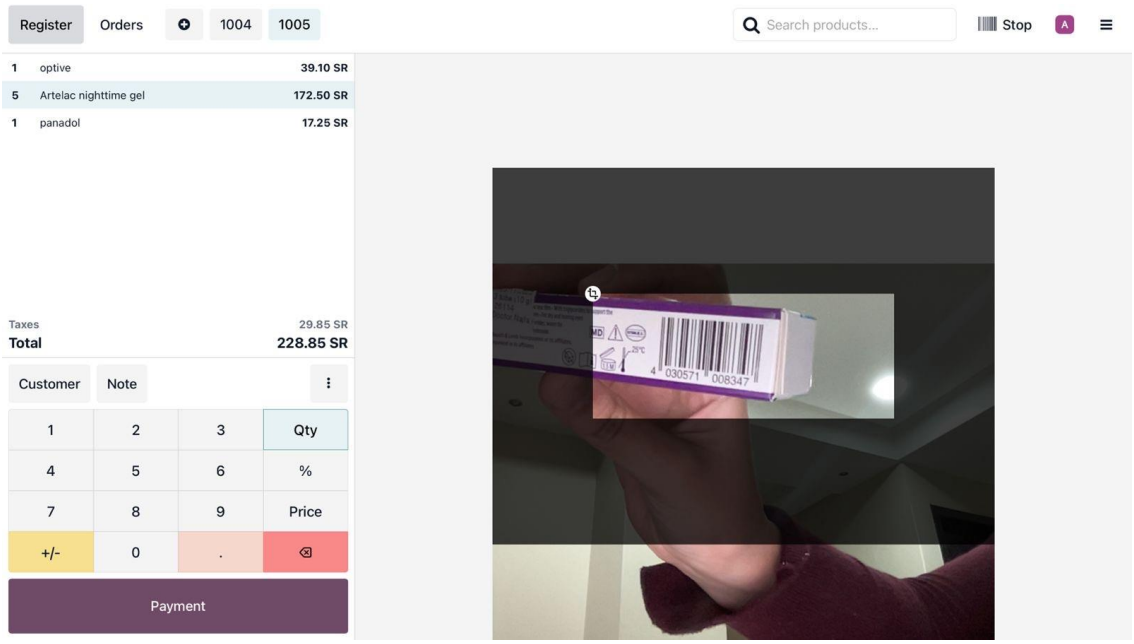


Figure 21Scan the barcode while

4.3.3 Payment and completion of the process.

After selecting products the customer is shown the total amount due along with payment options cash or card The transaction is then linked to the customer's account to process the order and issue an invoice.

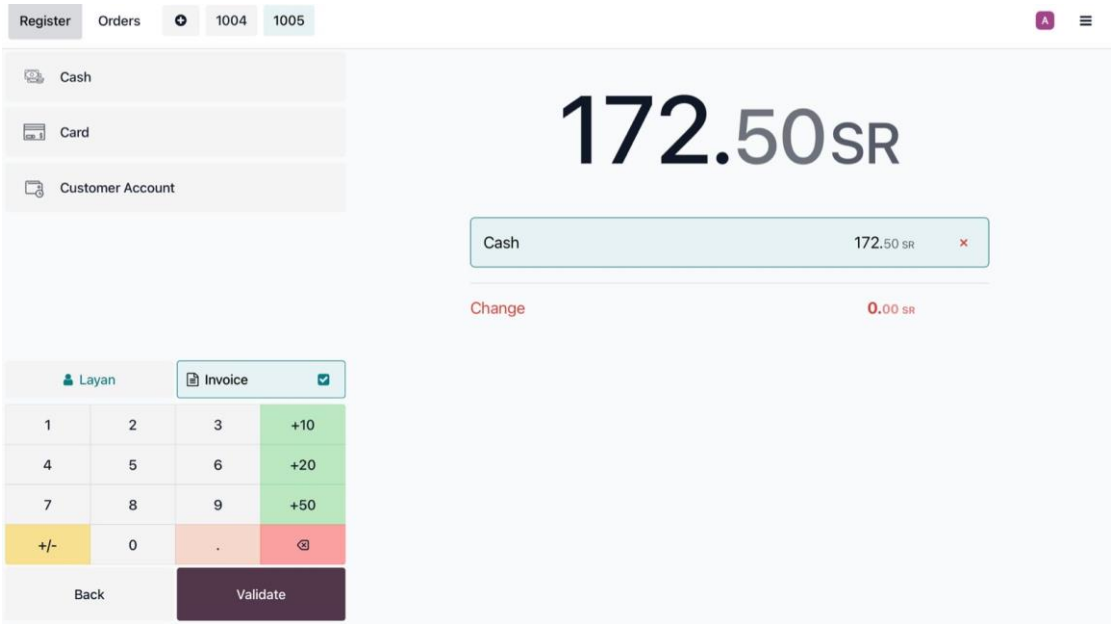


Figure 22Payment and completion of the process.

4.3.4 Invoice after payment.

The system displays a payment receipt with a QR code and a simplified tax invoice and the invoice can be sent directly to the customer email or printed. This step completes the process.

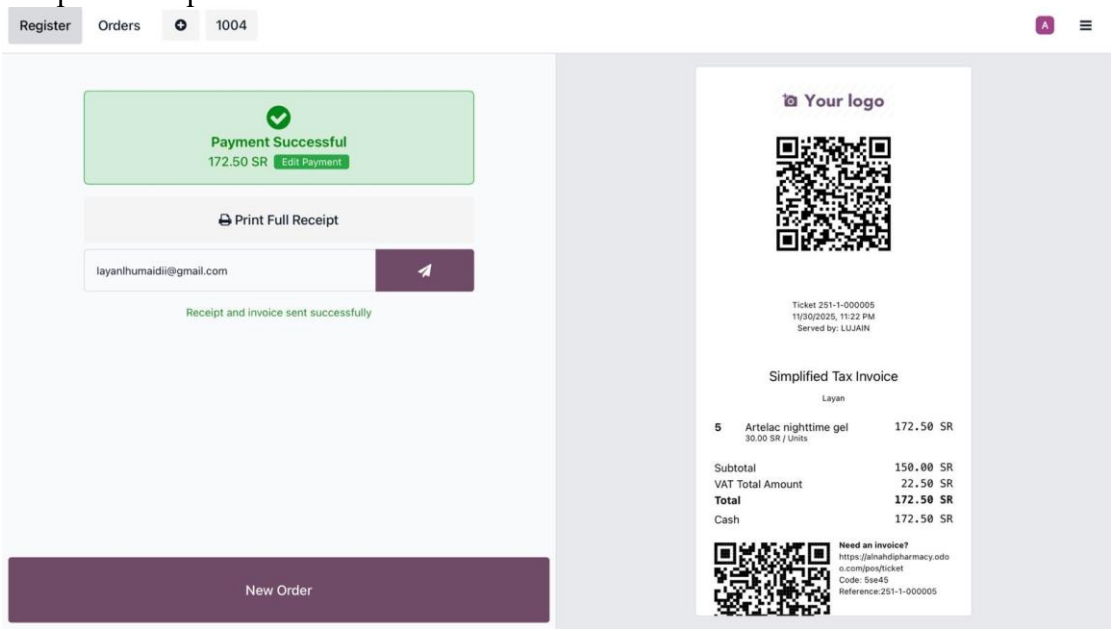
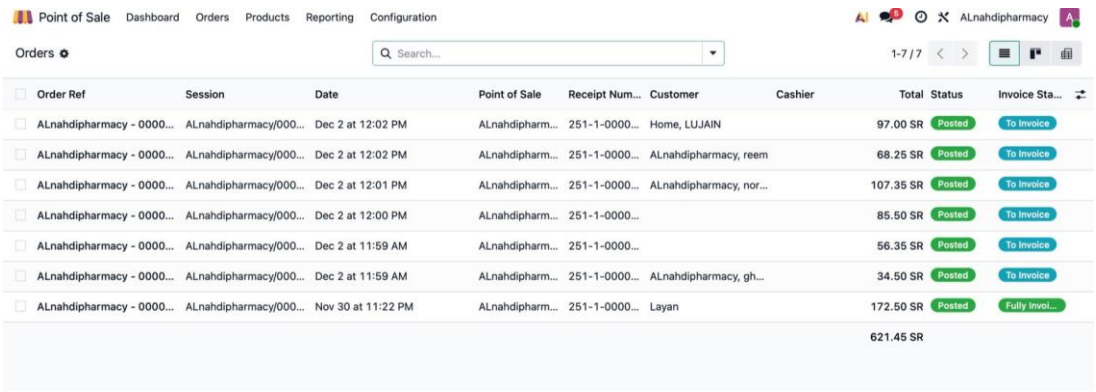


Figure 23Invoice after payment.

4.3.5 records of purchase transaction information with invoice.

After the purchases these records were entered into our invoices under orders.



4.3.6 Creating branches (Warehouses) in Odoo.

Several warehouses were created within the Odoo Inventory system to represent Nahdi Pharmacy branches in the proposed model so each branch has an independent stock location and allowing for product distribution between branches and tracking of available quantities at each branch separately this setup makes it easier for the system to suggest the nearest branch where items are available to the customer when the product is unavailable at their current branch.

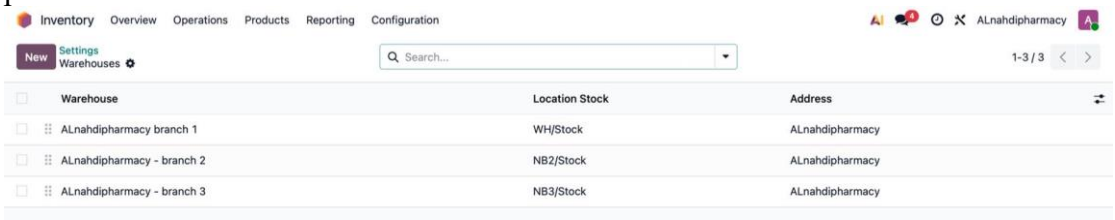


Figure 24Creating branches

4.3.7 Creating Storage Shelves Within the Warehouse.

I created internal storage locations (shelves) within the main Odoo warehouse such as shelf 1 by selecting the parent location WH/Stock and then choosing the location type Internal this step allows for organizing products within the warehouse and making it easier to accurately track their storage locations.

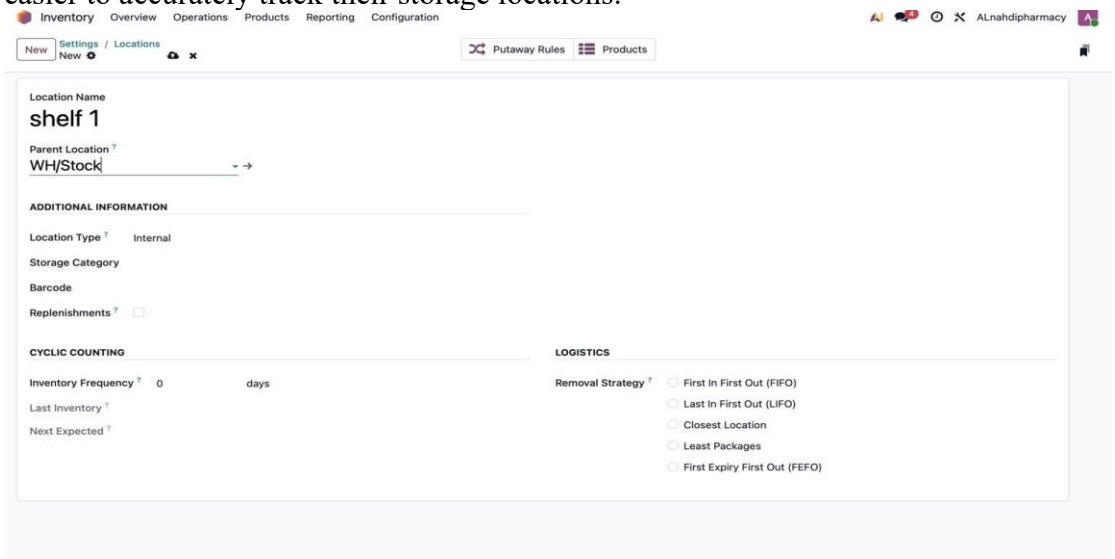


Figure 25Creating Storage Shelves

4.3.8 Displaying the locations of the storage shelves and facilities.

This image shows the storage shelf locations created within the warehouse shelf 1 and shelf 2 and shelf 3 and shelf 4 and 6 this organization helps improve inventory accuracy and speeds up product retrieval.

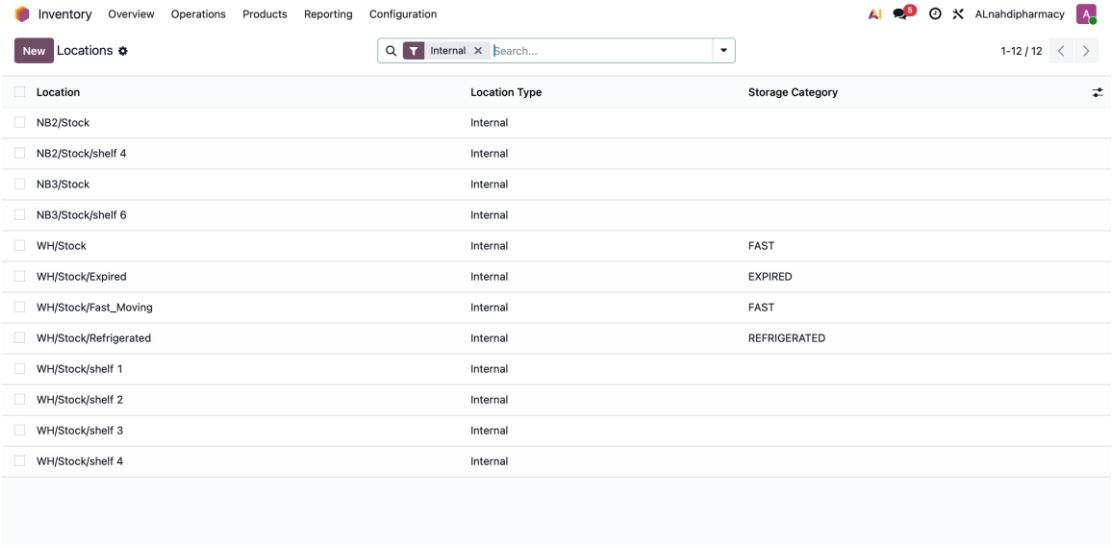


Figure 26Displaying the locations of the storage

4.3.6 Distributing the Vitamin D product on shelves.

The image illustrates how Vitamin D is distributed in warehouses with the quantity at each location identified and linked to specific shelves within branches the total stock and details for each location are shown and making it easy for pharmacists to quickly locate the product when searching for it.

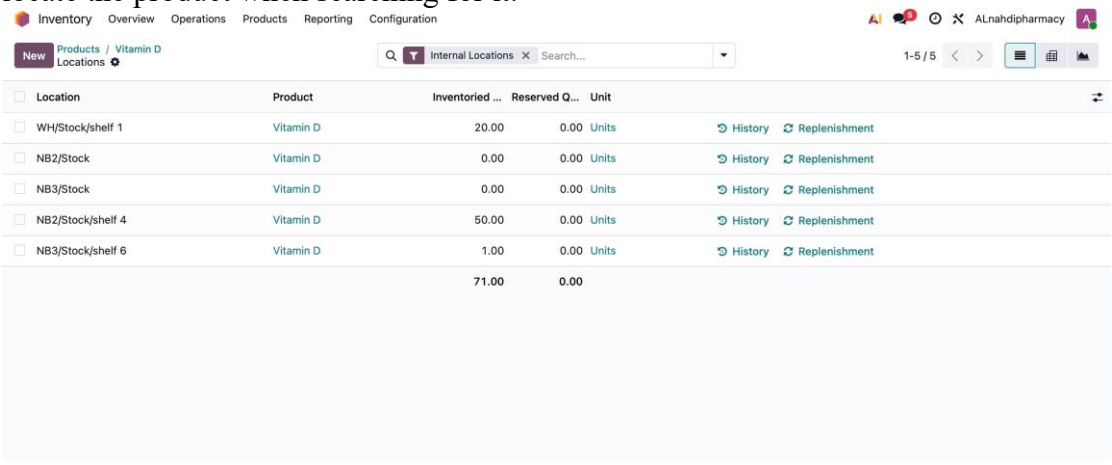


Figure 27Distributing the Vitamin D product on shelves.

4.4 CRM Complaint Process in Odoo

4.4.1 Customer Complaint Submission Form

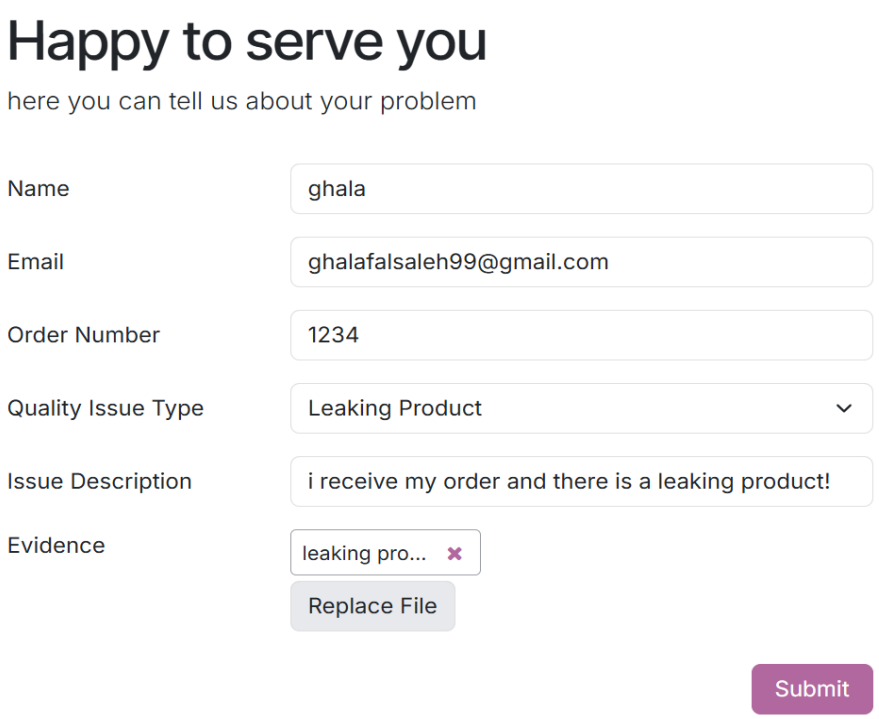


Figure 28Customer Complaint Submission Form

Alnahdi's website will have a unified online form for complaints, where customers will be able to directly put in their issue without calling/contacting agent on phone or leaving messages for them.
This online form greatly cuts down on handoffs and holes in communication. This is because it allows customers to enter all relevant information themselves in a single place. So, they can enter their name, email, order number, type of issue, description of the issue, and evidence.
When the form is submitted, the data will automatically go into the system which allows complaints to be registered faster and gives customers an overall better experience.

4.4.2 Instant Submission Feedback

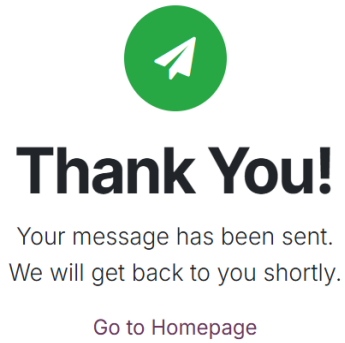


Figure 29Instant Submission Feedback

Once the customer submits a complaint form, the acknowledgment page will be displayed automatically by the system that the complaint has been submitted successfully.
The “Thank You” message lets the customer know that the request has been taken into the system. The support team will contact the customer shortly.

This auto-confirmation increases transparency and builds customer trust as the user knows that the complaint does get delivered immediately and doesn't need follow-ups or any manual check.

4.4.3 Complaint Records & Status Overview

Opportunity	Contact Name	Email	Quality Issue Type	Salesperson	Stage
ALnahdipharmacy's opportunity	layan	layan@gmail.com	damaged package	A LUJAIN	Closed / Resolved
ALnahdipharmacy's opportunity	norah	norah@gmail.]om	high temperature	A LUJAIN	Closed / Resolved
ALnahdipharmacy's opportunity	wed	wdalsalh04@gmail.com	fake product	A LUJAIN	Closed / Resolved
ALnahdipharmacy's opportunity	reem	reem@gmail.com	broken product	A LUJAIN	Closed / Resolved
ALnahdipharmacy's opportunity	nouf	nouf@gmail.com	wrong item	A LUJAIN	Closed / Resolved
ALnahdipharmacy's opportunity	ghala	ghalafalsaleh99@gmail....	leaking product	A LUJAIN	Closed / Resolved

Figure 30Complaint Records & Status Overview

As soon as the client completes the website form, a new complaint is automatically registered in the CRM system, without the aid of any agent.
The system validates the customer’s information received and logs it instantly in the CRM with correct Quality Issue Type as per customer selection in the form.

Complaints are all reflected in a table in which all the complaints get automatically classified into categories like damaged package, high temperature, fake product, broken product, wrong item, leaking product, etc.

With the automation to classify issues, each issue will get into the system clean and organized and ready for further processing.

4.4.4 Final Decision Notification to Customer

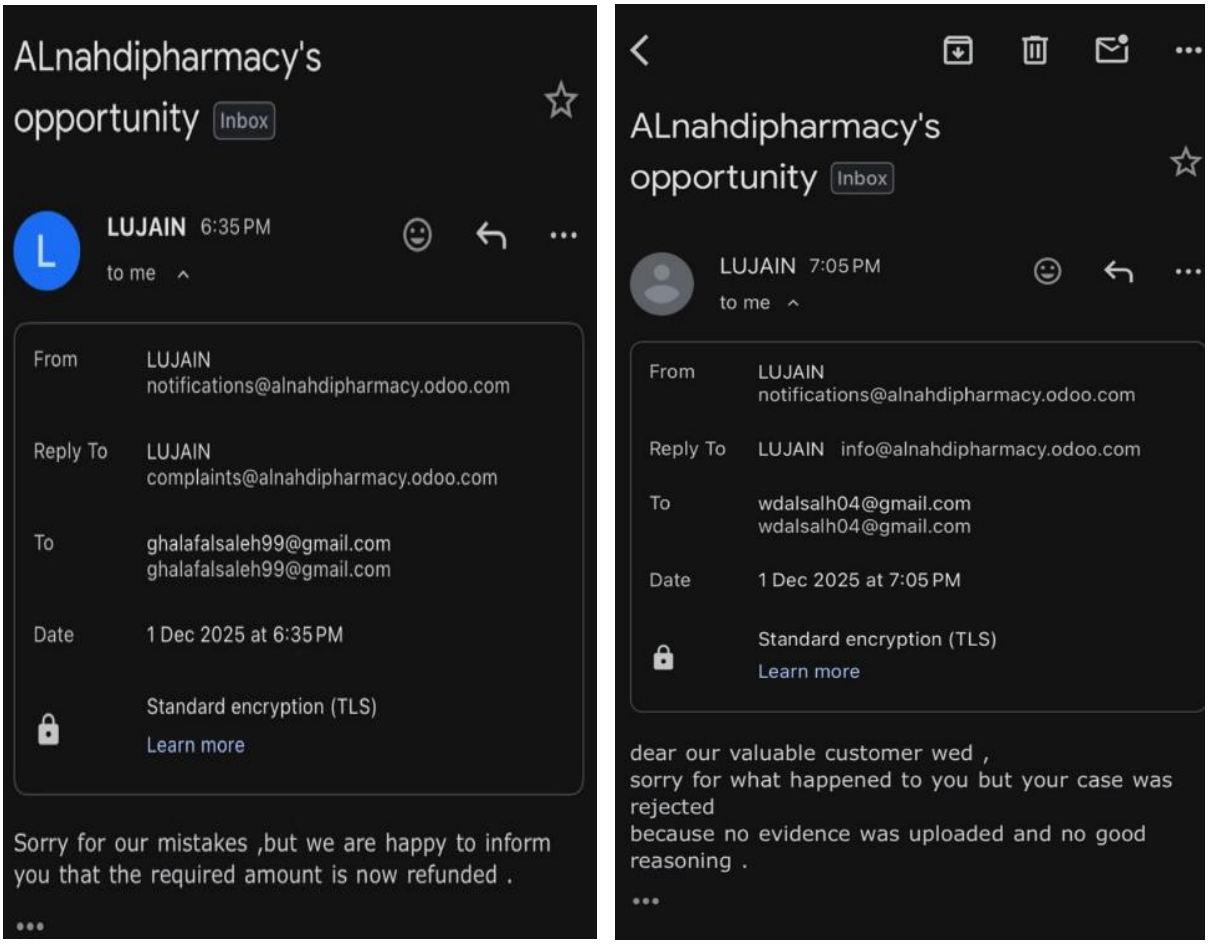


Figure 31Final Decision Notification to Customer

Once the complaint and proof have been analyzed, the system automatically determines its status – Approved or Rejected – the system sends an automated email to the customer.
An approval email tells the customer that the refund has been done
A rejection email informs the customer that their case has been rejected because of a lack of evidence or faulty reasoning.
When the email goes out, the CRM automatically updates to “Closed/Resolved”.
Which means, the complaint has completed its life cycle without requiring manual work.
Automation helps in keeping things consistent, fair, and also in informing the customers instantly.